THE UNIVERSITY OF SYDNEY

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CONFIDENTIAL EXAM PAPER

This paper is not to be removed from the exam venue

Electrical and Information Engineering

EXAMINATION

Semester 1 - Main, 2018

ELEC3607 Embedded Systems

EXAM WRITING TIME: 2 hours

READING TIME: 10 minutes

EXAM CONDITIONS:
This is a CLOSED book examination - no material permitted

During reading time - writing is not permitted at all

MATERIALS PERMITTED IN THE EXAM VENUE:
(No electronic aids are permitted e.g. laptops, phones)

Calculator - non-programmable

MATERIALS TO BE SUPPLIED TO STUDENTS:
1 x 12-page answer book

INSTRUCTIONS TO STUDENTS:
1. Please check your examination paper is complete (page numbers in footer of page) and indicate by signing below.

2. Please read each question carefully. The value of each question has been written next to the question. Note that questions are of unequal value. Total mark of this exam paper is 100.

3. This examination consists of four (4) SECTIONS (1, 2, 3, and 4).

4. ALL sections must be answered in the answer booklet provided.

5. Parts of the SAM3X datasheet and an ARM instruction set summary are provided at the end of this paper.

For Examiner Use Only

Q	Mark
1	
2	
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Total _____

Please tick the box to confirm that your examination paper is complete. ☐

Question 1 Peripherals (25 marks)

A data sheet for the SAM3X Watchdog Timer (WDT) peripheral is included at the end of this paper. In answering this question, you must give all constants for register addresses and values in hexadecimal.

a. (5 marks) We would like to program the WDT so that a processor reset occurs if a watchdog period of 10 seconds expires, irrespective of the processor state. The processor should not be reset upon restart. Assuming that the chip has just been reset and the slow clock is 32.768 kHz, give the names, addresses and values for all relevant registers needed to make it operational. Justify the values chosen for each field. Do not include any code in your answer.

b. (10 marks) In the C programming language, write a subroutine void setwdt(double p) which sets the watchdog timer to period p. Your answer should not use any library calls or include files.

c. (5 marks) In the C programming language, write a subroutine void reloadwdt() which will reload the watchdog timer. This function should be called periodically to avoid having a reset occur. For this subroutine, use symbolic names as used in CMSIS.

d. (5 marks) Give an example of a situation in which a watchdog timer would be used in an embedded system.

Question 2 ARM Assembly Language (25 marks)

An ARM instruction set summary is provided at the end of this paper.

a. (5 marks) Explain the difference between eor and eors instruction. Use an example to show why both forms are useful.

b. (5 marks) Explain using an example what the “ldr r3, [r7,#4]” instruction does.

c. (10 marks) The following is the assembly language generated by a C compiler.

```
        .type   mystery, %function
mystery:
        @ args = 0, pretend = 0, frame = 8
        @ frame_needed = 1, uses_anonymous_args = 0
        @ link register save eliminated.
        push    {r7}
        sub     sp, sp, #12
        add     r7, sp, #0
        str     r0, [r7, #4]
        str     r1, [r7]
        ldr     r2, [r7]
        ldr     r3, [r7, #4]
        cmp     r2, r3
        it      cs
        movcs   r3, r2
        mov     r0, r3
        adds    r7, r7, #12
        mov     sp, r7
        @ sp needed
        ldr     r7, [sp], #4
        bx      lr
```

Annotate each line of the function with comments to explain what each line does. Write an equivalent C function and explain what it computes.

d. (5 marks) Could the assembly language function above be executed using only the Thumb-2 instruction set? Explain how you arrived at your answer.

Question 3 Polling and Interrupts (25 marks)

a. (5 marks) Explain how polling can be used with the WDT to generate an accurate 2 second delay without resetting the SAM3X. You can assume that the chip has just been reset and the slow clock is 32.768 kHz. Give the names, addresses and values for all relevant registers which need to be programmed. Justify the values chosen for each field. Do not include any code in your answer.

b. (5 marks) In the C programming language, write a subroutine which will configure the WDT to generate an interrupt as close as possible to 0.1 seconds. Your answer should not allow the WDT to reset the microcontroller and you should use standard CMSIS constants.

c. (10 marks) We wish to develop an interrupt-based traffic light, which continually cycles through 5 seconds of green, 1 second of amber and 3 seconds of red. Corresponding LEDs are on I/O pins 10, 11 and 12 respectively, e.g. to make a light green, you should execute

```
digitalWrite(10, HIGH);  // green
digitalWrite(11, LOW);   // amber
digitalWrite(12, LOW);   // red
```

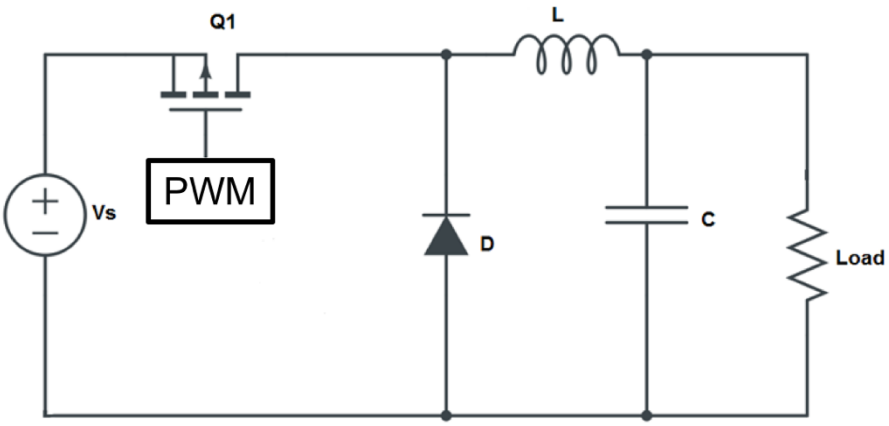
Draw a state diagram to illustrate your design and then write a C-based finite state machine based interrupt handler WDT_Handler(). Make sure you declare all variables needed.

d. (5 marks) Explain, using an example, the purpose of direct memory access. Give an example of a situation in which it is not advantageous over polling.

Question 4 Hardware Design (25 marks)

a. (5 marks) With the aid of a figure, explain ground bounce and give two techniques which can minimize this effect.

b. (5 marks) Explain the principle of operation of the power supply given in the figure below.



c. (5 marks) Why is quartz often used for oscillators? What physical principle is used?

d. (5 marks) In the context of interrupt handlers, explain what is a “race condition”. Use an example and also explain how it can be avoided.

e. (5 marks) A value of 0x3B is transmitted using a UART. Draw the waveform of the output assuming 9600 baud, 8 data bits, no parity and 1 stop bit. Label both the X and Y axes.

ARM® and Thumb®-2 Instruction Set
Quick Reference Card

Key to Tables			
`Rm {, <opsh>}`	See Table **Register, optionally shifted by constant**	`<reglist>`	A comma-separated list of registers, enclosed in braces { and }.
`<Operand2>`	See Table **Flexible Operand 2**. Shift and rotate are only available as part of Operand2.	`<reglist-PC>`	As `<reglist>`, must not include the PC.
`<fields>`	See Table **PSR fields**.	`<reglist+PC>`	As `<reglist>`, including the PC.
`<PSR>`	APSR (Application Program Status Register), CPSR (Current Processor Status Register), or SPSR (Saved Processor Status Register)	`<flags>`	Either `nzcvq` (ALU flags PSR[31:27]) or `g` (SIMD GE flags PSR[19:16])
`C*, V*`	Flag is unpredictable in Architecture v4 and earlier, unchanged in Architecture v5 and later.	§	See Table **ARM architecture versions**.
`<Rs|sh>`	Can be Rs or an immediate shift value. The values allowed for each shift type are the same as those shown in Table **Register, optionally shifted by constant**.	`+/-`	+ or –. (+ may be omitted.)
		`<iflags>`	Interrupt flags. One or more of `a`, `i`, `f` (abort, interrupt, fast interrupt).
`x,y`	B meaning half-register [15:0], or T meaning [31:16].	`<p_mode>`	See Table **Processor Modes**
`<imm8m>`	ARM: a 32-bit constant, formed by right-rotating an 8-bit value by an even number of bits.	`SPm`	SP for the processor mode specified by `<p_mode>`
	Thumb: a 32-bit constant, formed by left-shifting an 8-bit value by any number of bits, or a bit pattern of one of the forms 0xXYXYXYXY, 0x00XY00XY or 0xXY00XY00.	`<lsb>`	Least significant bit of bitfield.
		`<width>`	Width of bitfield. `<width>` + `<lsb>` must be <= 32.
`<prefix>`	See Table **Prefixes for Parallel instructions**	`{X}`	RsX is Rs rotated 16 bits if X present. Otherwise, RsX is Rs.
`{IA|IB|DA|DB}`	Increment After, Increment Before, Decrement After, or Decrement Before. `IB` and `DA` are not available in Thumb state. If omitted, defaults to `IA`.	`{!}`	Updates base register after data transfer if ! present (pre-indexed).
		`{S}`	Updates condition flags if S present.
`<size>`	`B`, `SB`, `H`, or `SH`, meaning Byte, Signed Byte, Halfword, and Signed Halfword respectively. `SB` and `SH` are not available in `STR` instructions.	`{T}`	User mode privilege if T present.
		`{R}`	Rounds result to nearest if R present, otherwise truncates result.

Operation		§	Assembler	S updates	Action	Notes
Add	Add		`ADD{S} Rd, Rn, <Operand2>`	N Z C V	Rd := Rn + Operand2	N
	with carry		`ADC{S} Rd, Rn, <Operand2>`	N Z C V	Rd := Rn + Operand2 + Carry	N
	wide	T2	`ADD Rd, Rn, #<imm12>`		Rd := Rn + imm12, imm12 range 0-4095	T, P
	saturating {doubled}	5E	`Q{D}ADD Rd, Rm, Rn`		Rd := SAT(Rm + Rn) doubled: Rd := SAT(Rm + SAT(Rn * 2))	Q
Address	Form PC-relative address		`ADR Rd, <label>`		Rd := <label>, for <label> range from current instruction see Note L	N, L
Subtract	Subtract		`SUB{S} Rd, Rn, <Operand2>`	N Z C V	Rd := Rn – Operand2	N
	with carry		`SBC{S} Rd, Rn, <Operand2>`	N Z C V	Rd := Rn – Operand2 – NOT(Carry)	N
	wide	T2	`SUB Rd, Rn, #<imm12>`		Rd := Rn – imm12, imm12 range 0-4095	T, P
	reverse subtract		`RSB{S} Rd, Rn, <Operand2>`	N Z C V	Rd := Operand2 – Rn	N
	reverse subtract with carry		`RSC{S} Rd, Rn, <Operand2>`	N Z C V	Rd := Operand2 – Rn – NOT(Carry)	A
	saturating {doubled}	5E	`Q{D}SUB Rd, Rm, Rn`		Rd := SAT(Rm – Rn) doubled: Rd := SAT(Rm – SAT(Rn * 2))	Q
	Exception return without stack		`SUBS PC, LR, #<imm8>`	N Z C V	PC = LR – imm8, CPSR = SPSR(current mode), imm8 range 0-255.	
Parallel arithmetic	Halfword-wise addition	6	`<prefix>ADD16 Rd, Rn, Rm`		Rd[31:16] := Rn[31:16] + Rm[31:16], Rd[15:0] := Rn[15:0] + Rm[15:0]	G
	Halfword-wise subtraction	6	`<prefix>SUB16 Rd, Rn, Rm`		Rd[31:16] := Rn[31:16] – Rm[31:16], Rd[15:0] := Rn[15:0] – Rm[15:0]	G
	Byte-wise addition	6	`<prefix>ADD8 Rd, Rn, Rm`		Rd[31:24] := Rn[31:24] + Rm[31:24], Rd[23:16] := Rn[23:16] + Rm[23:16], Rd[15:8] := Rn[15:8] + Rm[15:8], Rd[7:0] := Rn[7:0] + Rm[7:0]	G
	Byte-wise subtraction	6	`<prefix>SUB8 Rd, Rn, Rm`		Rd[31:24] := Rn[31:24] – Rm[31:24], Rd[23:16] := Rn[23:16] – Rm[23:16], Rd[15:8] := Rn[15:8] – Rm[15:8], Rd[7:0] := Rn[7:0] – Rm[7:0]	G
	Halfword-wise exchange, add, subtract	6	`<prefix>ASX Rd, Rn, Rm`		Rd[31:16] := Rn[31:16] + Rm[15:0], Rd[15:0] := Rn[15:0] – Rm[31:16]	G
	Halfword-wise exchange, subtract, add	6	`<prefix>SAX Rd, Rn, Rm`		Rd[31:16] := Rn[31:16] – Rm[15:0], Rd[15:0] := Rn[15:0] + Rm[31:16]	G
	Unsigned sum of absolute differences	6	`USAD8 Rd, Rm, Rs`		Rd := Abs(Rm[31:24] – Rs[31:24]) + Abs(Rm[23:16] – Rs[23:16]) + Abs(Rm[15:8] – Rs[15:8]) + Abs(Rm[7:0] – Rs[7:0])	
	and accumulate	6	`USADA8 Rd, Rm, Rs, Rn`		Rd := Rn + Abs(Rm[31:24] – Rs[31:24]) + Abs(Rm[23:16] – Rs[23:16]) + Abs(Rm[15:8] – Rs[15:8]) + Abs(Rm[7:0] – Rs[7:0])	
Saturate	Signed saturate word, right shift	6	`SSAT Rd, #<sat>, Rm{, ASR <sh>}`		Rd := SignedSat((Rm ASR sh), sat). `<sat>` range 1-32, `<sh>` range 1-31.	Q, R
	Signed saturate word, left shift	6	`SSAT Rd, #<sat>, Rm{, LSL <sh>}`		Rd := SignedSat((Rm LSL sh), sat). `<sat>` range 1-32, `<sh>` range 0-31.	Q
	Signed saturate two halfwords	6	`SSAT16 Rd, #<sat>, Rm`		Rd[31:16] := SignedSat(Rm[31:16], sat), Rd[15:0] := SignedSat(Rm[15:0], sat). `<sat>` range 1-16.	Q
	Unsigned saturate word, right shift	6	`USAT Rd, #<sat>, Rm{, ASR <sh>}`		Rd := UnsignedSat((Rm ASR sh), sat). `<sat>` range 0-31, `<sh>` range 1-31.	Q, R
	Unsigned saturate word, left shift	6	`USAT Rd, #<sat>, Rm{, LSL <sh>}`		Rd := UnsignedSat((Rm LSL sh), sat). `<sat>` range 0-31, `<sh>` range 0-31.	Q
	Unsigned saturate two halfwords	6	`USAT16 Rd, #<sat>, Rm`		Rd[31:16] := UnsignedSat(Rm[31:16], sat), Rd[15:0] := UnsignedSat(Rm[15:0], sat). `<sat>` range 0-15.	Q

ARM and Thumb-2 Instruction Set
Quick Reference Card

Operation		§	Assembler	S updates	Action	Notes
Multiply	Multiply		`MUL{S} Rd, Rm, Rs`	N Z C*	Rd := (Rm * Rs)[31:0] (If Rs is Rd, S can be used in Thumb-2)	N, S
	and accumulate		`MLA{S} Rd, Rm, Rs, Rn`	N Z C*	Rd := (Rn + (Rm * Rs))[31:0]	S
	and subtract	T2	`MLS Rd, Rm, Rs, Rn`		Rd := (Rn – (Rm * Rs))[31:0]	
	unsigned long		`UMULL{S} RdLo, RdHi, Rm, Rs`	N Z C* V*	RdHi,RdLo := unsigned(Rm * Rs)	S
	unsigned accumulate long		`UMLAL{S} RdLo, RdHi, Rm, Rs`	N Z C* V*	RdHi,RdLo := unsigned(RdHi,RdLo + Rm * Rs)	S
	unsigned double accumulate long	6	`UMAAL RdLo, RdHi, Rm, Rs`		RdHi,RdLo := unsigned(RdHi + RdLo + Rm * Rs)	
	Signed multiply long		`SMULL{S} RdLo, RdHi, Rm, Rs`	N Z C* V*	RdHi,RdLo := signed(Rm * Rs)	S
	and accumulate long		`SMLAL{S} RdLo, RdHi, Rm, Rs`	N Z C* V*	RdHi,RdLo := signed(RdHi,RdLo + Rm * Rs)	S
	16 * 16 bit	5E	`SMULxy Rd, Rm, Rs`		Rd := Rm[x] * Rs[y]	
	32 * 16 bit	5E	`SMULWy Rd, Rm, Rs`		Rd := (Rm * Rs[y])[47:16]	
	16 * 16 bit and accumulate	5E	`SMLAxy Rd, Rm, Rs, Rn`		Rd := Rn + Rm[x] * Rs[y]	Q
	32 * 16 bit and accumulate	5E	`SMLAWy Rd, Rm, Rs, Rn`		Rd := Rn + (Rm * Rs[y])[47:16]	Q
	16 * 16 bit and accumulate long	5E	`SMLALxy RdLo, RdHi, Rm, Rs`		RdHi,RdLo := RdHi,RdLo + Rm[x] * Rs[y]	
	Dual signed multiply, add	6	`SMUAD{X} Rd, Rm, Rs`		Rd := Rm[15:0] * RsX[15:0] + Rm[31:16] * RsX[31:16]	Q
	and accumulate	6	`SMLAD{X} Rd, Rm, Rs, Rn`		Rd := Rn + Rm[15:0] * RsX[15:0] + Rm[31:16] * RsX[31:16]	Q
	and accumulate long	6	`SMLALD{X} RdLo, RdHi, Rm, Rs`		RdHi,RdLo := RdHi,RdLo + Rm[15:0] * RsX[15:0] + Rm[31:16] * RsX[31:16]	
	Dual signed multiply, subtract	6	`SMUSD{X} Rd, Rm, Rs`		Rd := Rm[15:0] * RsX[15:0] – Rm[31:16] * RsX[31:16]	Q
	and accumulate	6	`SMLSD{X} Rd, Rm, Rs, Rn`		Rd := Rn + Rm[15:0] * RsX[15:0] – Rm[31:16] * RsX[31:16]	Q
	and accumulate long	6	`SMLSLD{X} RdLo, RdHi, Rm, Rs`		RdHi,RdLo := RdHi,RdLo + Rm[15:0] * RsX[15:0] – Rm[31:16] * RsX[31:16]	
	Signed top word multiply	6	`SMMUL{R} Rd, Rm, Rs`		Rd := (Rm * Rs)[63:32]	
	and accumulate	6	`SMMLA{R} Rd, Rm, Rs, Rn`		Rd := Rn + (Rm * Rs)[63:32]	
	and subtract	6	`SMMLS{R} Rd, Rm, Rs, Rn`		Rd := Rn – (Rm * Rs)[63:32]	
	with internal 40-bit accumulate	XS	`MIA Ac, Rm, Rs`		Ac := Ac + Rm * Rs	
	packed halfword	XS	`MIAPH Ac, Rm, Rs`		Ac := Ac + Rm[15:0] * Rs[15:0] + Rm[31:16] * Rs[31:16]	
	halfword	XS	`MIAxy Ac, Rm, Rs`		Ac := Ac + Rm[x] * Rs[y]	
Divide	Signed or Unsigned	RM	`<op> Rd, Rn, Rm`		Rd := Rn / Rm <op> is `SDIV` (signed) or `UDIV` (unsigned)	T
Move data	Move		`MOV{S} Rd, <Operand2>`	N Z C	Rd := Operand2 See also Shift instructions	N
	not		`MVN{S} Rd, <Operand2>`	N Z C	Rd := 0xFFFFFFFF EOR Operand2	N
	top	T2	`MOVT Rd, #<imm16>`		Rd[31:16] := imm16, Rd[15:0] unaffected, imm16 range 0-65535	
	wide	T2	`MOV Rd, #<imm16>`		Rd[15:0] := imm16, Rd[31:16] = 0, imm16 range 0-65535	
	40-bit accumulator to register	XS	`MRA RdLo, RdHi, Ac`		RdLo := Ac[31:0], RdHi := Ac[39:32]	
	register to 40-bit accumulator	XS	`MAR Ac, RdLo, RdHi`		Ac[31:0] := RdLo, Ac[39:32] := RdHi[7:0]	
Shift	Arithmetic shift right		`ASR{S} Rd, Rm, <Rs|sh>`	N Z C	Rd := ASR(Rm, Rs|sh) Same as `MOV{S} Rd, Rm, ASR <Rs|sh>`	N
	Logical shift left		`LSL{S} Rd, Rm, <Rs|sh>`	N Z C	Rd := LSL(Rm, Rs|sh) Same as `MOV{S} Rd, Rm, LSL <Rs|sh>`	N
	Logical shift right		`LSR{S} Rd, Rm, <Rs|sh>`	N Z C	Rd := LSR(Rm, Rs|sh) Same as `MOV{S} Rd, Rm, LSR <Rs|sh>`	N
	Rotate right		`ROR{S} Rd, Rm, <Rs|sh>`	N Z C	Rd := ROR(Rm, Rs|sh) Same as `MOV{S} Rd, Rm, ROR <Rs|sh>`	N
	Rotate right with extend		`RRX{S} Rd, Rm`	N Z C	Rd := RRX(Rm) Same as `MOV{S} Rd, Rm, RRX`	
Count leading zeros		5	`CLZ Rd, Rm`		Rd := number of leading zeros in Rm	
Compare	Compare		`CMP Rn, <Operand2>`	N Z C V	Update CPSR flags on Rn – Operand2	N
	negative		`CMN Rn, <Operand2>`	N Z C V	Update CPSR flags on Rn + Operand2	N
Logical	Test		`TST Rn, <Operand2>`	N Z C	Update CPSR flags on Rn AND Operand2	N
	Test equivalence		`TEQ Rn, <Operand2>`	N Z C	Update CPSR flags on Rn EOR Operand2	
	AND		`AND{S} Rd, Rn, <Operand2>`	N Z C	Rd := Rn AND Operand2	N
	EOR		`EOR{S} Rd, Rn, <Operand2>`	N Z C	Rd := Rn EOR Operand2	N
	ORR		`ORR{S} Rd, Rn, <Operand2>`	N Z C	Rd := Rn OR Operand2	N
	ORN	T2	`ORN{S} Rd, Rn, <Operand2>`	N Z C	Rd := Rn OR NOT Operand2	T
	Bit Clear		`BIC{S} Rd, Rn, <Operand2>`	N Z C	Rd := Rn AND NOT Operand2	N

ARM and Thumb-2 Instruction Set
Quick Reference Card

Operation		§	Assembler	Action	Notes
Bit field	Bit Field Clear	T2	`BFC Rd, #<lsb>, #<width>`	Rd[(width+lsb–1):lsb] := 0, other bits of Rd unaffected	
	Bit Field Insert	T2	`BFI Rd, Rn, #<lsb>, #<width>`	Rd[(width+lsb–1):lsb] := Rn[(width-1):0], other bits of Rd unaffected	
	Signed Bit Field Extract	T2	`SBFX Rd, Rn, #<lsb>, #<width>`	Rd[(width–1):0] = Rn[(width+lsb–1):lsb], Rd[31:width] = Replicate(Rn[width+lsb–1])	
	Unsigned Bit Field Extract	T2	`UBFX Rd, Rn, #<lsb>, #<width>`	Rd[(width–1):0] = Rn[(width+lsb–1):lsb], Rd[31:width] = Replicate(0)	
Pack	Pack halfword bottom + top	6	`PKHBT Rd, Rn, Rm{, LSL #<sh>}`	Rd[15:0] := Rn[15:0], Rd[31:16] := (Rm LSL sh)[31:16]. sh 0-31.	
	Pack halfword top + bottom	6	`PKHTB Rd, Rn, Rm{, ASR #<sh>}`	Rd[31:16] := Rn[31:16], Rd[15:0] := (Rm ASR sh)[15:0]. sh 1-32.	
Signed extend	Halfword to word	6	`SXTH Rd, Rm{, ROR #<sh>}`	Rd[31:0] := SignExtend((Rm ROR (8 * sh))[15:0]). sh 0-3.	N
	Two bytes to halfwords	6	`SXTB16 Rd, Rm{, ROR #<sh>}`	Rd[31:16] := SignExtend((Rm ROR (8 * sh))[23:16]), Rd[15:0] := SignExtend((Rm ROR (8 * sh))[7:0]). sh 0-3.	
	Byte to word	6	`SXTB Rd, Rm{, ROR #<sh>}`	Rd[31:0] := SignExtend((Rm ROR (8 * sh))[7:0]). sh 0-3.	N
Unsigned extend	Halfword to word	6	`UXTH Rd, Rm{, ROR #<sh>}`	Rd[31:0] := ZeroExtend((Rm ROR (8 * sh))[15:0]). sh 0-3.	N
	Two bytes to halfwords	6	`UXTB16 Rd, Rm{, ROR #<sh>}`	Rd[31:16] := ZeroExtend((Rm ROR (8 * sh))[23:16]), Rd[15:0] := ZeroExtend((Rm ROR (8 * sh))[7:0]). sh 0-3.	
	Byte to word	6	`UXTB Rd, Rm{, ROR #<sh>}`	Rd[31:0] := ZeroExtend((Rm ROR (8 * sh))[7:0]). sh 0-3.	N
Signed extend with add	Halfword to word, add	6	`SXTAH Rd, Rn, Rm{, ROR #<sh>}`	Rd[31:0] := Rn[31:0] + SignExtend((Rm ROR (8 * sh))[15:0]). sh 0-3.	
	Two bytes to halfwords, add	6	`SXTAB16 Rd, Rn, Rm{, ROR #<sh>}`	Rd[31:16] := Rn[31:16] + SignExtend((Rm ROR (8 * sh))[23:16]), Rd[15:0] := Rn[15:0] + SignExtend((Rm ROR (8 * sh))[7:0]). sh 0-3.	
	Byte to word, add	6	`SXTAB Rd, Rn, Rm{, ROR #<sh>}`	Rd[31:0] := Rn[31:0] + SignExtend((Rm ROR (8 * sh))[7:0]). sh 0-3.	
Unsigned extend with add	Halfword to word, add	6	`UXTAH Rd, Rn, Rm{, ROR #<sh>}`	Rd[31:0] := Rn[31:0] + ZeroExtend((Rm ROR (8 * sh))[15:0]). sh 0-3.	
	Two bytes to halfwords, add	6	`UXTAB16 Rd, Rn, Rm{, ROR #<sh>}`	Rd[31:16] := Rn[31:16] + ZeroExtend((Rm ROR (8 * sh))[23:16]), Rd[15:0] := Rn[15:0] + ZeroExtend((Rm ROR (8 * sh))[7:0]). sh 0-3.	
	Byte to word, add	6	`UXTAB Rd, Rn, Rm{, ROR #<sh>}`	Rd[31:0] := Rn[31:0] + ZeroExtend((Rm ROR (8 * sh))[7:0]). sh 0-3.	
Reverse	Bits in word	T2	`RBIT Rd, Rm`	For (i = 0; i < 32; i++) : Rd[i] = Rm[31– i]	
	Bytes in word	6	`REV Rd, Rm`	Rd[31:24] := Rm[7:0], Rd[23:16] := Rm[15:8], Rd[15:8] := Rm[23:16], Rd[7:0] := Rm[31:24]	N
	Bytes in both halfwords	6	`REV16 Rd, Rm`	Rd[15:8] := Rm[7:0], Rd[7:0] := Rm[15:8], Rd[31:24] := Rm[23:16], Rd[23:16] := Rm[31:24]	N
	Bytes in low halfword, sign extend	6	`REVSH Rd, Rm`	Rd[15:8] := Rm[7:0], Rd[7:0] := Rm[15:8], Rd[31:16] := Rm[7] * &FFFF	N
Select	Select bytes	6	`SEL Rd, Rn, Rm`	Rd[7:0] := Rn[7:0] if GE[0] = 1, else Rd[7:0] := Rm[7:0] Bits[15:8], [23:16], [31:24] selected similarly by GE[1], GE[2], GE[3]	
If-Then	If-Then	T2	`IT{pattern} {cond}`	Makes up to four following instructions conditional, according to pattern. pattern is a string of up to three letters. Each letter can be T (Then) or E (Else). The first instruction after IT has condition cond. The following instructions have condition cond if the corresponding letter is T, or the inverse of cond if the corresponding letter is E. See Table **Condition Field** for available condition codes.	T, U
Branch	Branch		`B <label>`	PC := label. label is this instruction ±32MB (T2: ±16MB, T: –252 - +256B)	N, B
	with link		`BL <label>`	LR := address of next instruction, PC := label. label is this instruction ±32MB (T2: ±16MB).	
	and exchange	4T	`BX Rm`	PC := Rm. Target is Thumb if Rm[0] is 1, ARM if Rm[0] is 0.	N
	with link and exchange (1)	5T	`BLX <label>`	LR := address of next instruction, PC := label. Change instruction set. label is this instruction ±32MB (T2: ±16MB).	C
	with link and exchange (2)	5	`BLX Rm`	LR := address of next instruction, PC := Rm[31:1]. Change to Thumb if Rm[0] is 1, to ARM if Rm[0] is 0.	N
	and change to Jazelle state	5J	`BXJ Rm`	Change to Jazelle state if available	
	Compare, branch if (non) zero	T2	`CB{N}Z Rn,<label>`	If Rn {== or !=} 0 then PC := label. label is (this instruction + 4-130).	N,T,U
	Table Branch Byte	T2	`TBB [Rn, Rm]`	PC = PC + ZeroExtend(Memory(Rn + Rm, 1) << 1). Branch range 4-512. Rn can be PC.	T, U
	Table Branch Halfword	T2	`TBH [Rn, Rm, LSL #1]`	PC = PC + ZeroExtend(Memory(Rn + Rm << 1, 2) << 1). Branch range 4-131072. Rn can be PC.	T, U
Move to or from PSR	PSR to register		`MRS Rd, <PSR>`	Rd := PSR	
	register flags to APSR flags		`MSR APSR_<flags>, Rm`	APSR_<flags> := Rm	
	immediate flags to APSR flags		`MSR APSR_<flags>, #<imm8m>`	APSR_<flags> := immed_8r	
	register to PSR		`MSR <PSR>_<fields>, Rm`	PSR := Rm (selected bytes only)	
	immediate to PSR		`MSR <PSR>_<fields>, #<imm8m>`	PSR := immed_8r (selected bytes only)	
Processor state change	Change processor state	6	`CPSID <iflags> {, #<p_mode>}`	Disable specified interrupts, optional change mode.	U, N
		6	`CPSIE <iflags> {, #<p_mode>}`	Enable specified interrupts, optional change mode.	U, N
	Change processor mode	6	`CPS #<p_mode>`		U
	Set endianness	6	`SETEND <endianness>`	Sets endianness for loads and stores. `<endianness>` can be `BE` (Big Endian) or `LE` (Little Endian).	U, N

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Single data item loads and stores		§	Assembler	Action if <op> is LDR	Action if <op> is STR	Notes
Load or store word, byte or halfword	Immediate offset		`<op>{size}{T} Rd, [Rn {, #<offset>}]{!}`	Rd := [address, size]	[address, size] := Rd	1, N
	Post-indexed, immediate		`<op>{size}{T} Rd, [Rn], #<offset>`	Rd := [address, size]	[address, size] := Rd	2
	Register offset		`<op>{size} Rd, [Rn, +/-Rm {, <opsh>}]{!}`	Rd := [address, size]	[address, size] := Rd	3, N
	Post-indexed, register		`<op>{size}{T} Rd, [Rn], +/-Rm {, <opsh>}`	Rd := [address, size]	[address, size] := Rd	4
	PC-relative		`<op>{size} Rd, <label>`	Rd := [label, size]	Not available	5, N
Load or store doubleword	Immediate offset	5E	`<op>D Rd1, Rd2, [Rn {, #<offset>}]{!}`	Rd1 := [address], Rd2 := [address + 4]	[address] := Rd1, [address + 4] := Rd2	6, 9
	Post-indexed, immediate	5E	`<op>D Rd1, Rd2, [Rn], #<offset>`	Rd1 := [address], Rd2 := [address + 4]	[address] := Rd1, [address + 4] := Rd2	6, 9
	Register offset	5E	`<op>D Rd1, Rd2, [Rn, +/-Rm {, <opsh>}]{!}`	Rd1 := [address], Rd2 := [address + 4]	[address] := Rd1, [address + 4] := Rd2	7, 9
	Post-indexed, register	5E	`<op>D Rd1, Rd2, [Rn], +/-Rm {, <opsh>}`	Rd1 := [address], Rd2 := [address + 4]	[address] := Rd1, [address + 4] := Rd2	7, 9
	PC-relative	5E	`<op>D Rd1, Rd2, <label>`	Rd1 := [label], Rd2 := [label + 4]	Not available	8, 9

Preload data or instruction		§(PLD)	§(PLI)	§(PLDW)	Assembler	Action if <op> is PLD	Action if <op> is PLI	Action if <op> is PLDW	Notes
	Immediate offset	5E	7	7MP	`<op> [Rn {, #<offset>}]`	Preload [address, 32] (data)	Preload [address, 32] (instruction)	Preload to Write [address, 32] (data)	1, C
	Register offset	5E	7	7MP	`<op> [Rn, +/-Rm {, <opsh>}]`	Preload [address, 32] (data)	Preload [address, 32] (instruction)	Preload to Write [address, 32] (data)	3, C
	PC-relative	5E	7		`<op> <label>`	Preload [label, 32] (data)	Preload [label, 32] (instruction)		5, C

Other memory operations		§	Assembler	Action	Notes
Load multiple	Block data load		`LDM{IA|IB|DA|DB} Rn{!}, <reglist-PC>`	Load list of registers from [Rn]	N, I
	return (and exchange)		`LDM{IA|IB|DA|DB} Rn{!}, <reglist+PC>`	Load registers, PC := [address][31:1] (§ 5T: Change to Thumb if [address][0] is 1)	I
	and restore CPSR		`LDM{IA|IB|DA|DB} Rn{!}, <reglist+PC>^`	Load registers, branch (§ 5T: and exchange), CPSR := SPSR. Exception modes only.	I
	User mode registers		`LDM{IA|IB|DA|DB} Rn, <reglist-PC>^`	Load list of User mode registers from [Rn]. Privileged modes only.	I
Pop			`POP <reglist>`	Canonical form of `LDM SP!, <reglist>`	N
Load exclusive	Semaphore operation	6	`LDREX Rd, [Rn]`	Rd := [Rn], tag address as exclusive access. Outstanding tag set if not shared address. Rd, Rn not PC.	
	Halfword or Byte	6K	`LDREX{H|B} Rd, [Rn]`	Rd[15:0] := [Rn] or Rd[7:0] := [Rn], tag address as exclusive access. Outstanding tag set if not shared address. Rd, Rn not PC.	
	Doubleword	6K	`LDREXD Rd1, Rd2, [Rn]`	Rd1 := [Rn], Rd2 := [Rn+4], tag addresses as exclusive access Outstanding tags set if not shared addresses. Rd1, Rd2, Rn not PC.	9
Store multiple	Push, or Block data store		`STM{IA|IB|DA|DB} Rn{!}, <reglist>`	Store list of registers to [Rn]	N, I
	User mode registers		`STM{IA|IB|DA|DB} Rn{!}, <reglist>^`	Store list of User mode registers to [Rn]. Privileged modes only.	I
Push			`PUSH <reglist>`	Canonical form of `STMDB SP!, <reglist>`	N
Store exclusive	Semaphore operation	6	`STREX Rd, Rm, [Rn]`	If allowed, [Rn] := Rm, clear exclusive tag, Rd := 0. Else Rd := 1. Rd, Rm, Rn not PC.	
	Halfword or Byte	6K	`STREX{H|B} Rd, Rm, [Rn]`	If allowed, [Rn] := Rm[15:0] or [Rn] := Rm[7:0], clear exclusive tag, Rd := 0. Else Rd := 1 Rd, Rm, Rn not PC.	
	Doubleword	6K	`STREXD Rd, Rm1, Rm2, [Rn]`	If allowed, [Rn] := Rm1, [Rn+4] := Rm2, clear exclusive tags, Rd := 0. Else Rd := 1 Rd, Rm1, Rm2, Rn not PC.	10
Clear exclusive		6K	`CLREX`	Clear local processor exclusive tag	C

Notes: availability and range of options for Load, Store, and Preload operations					
Note	**ARM Word, B, D**	**ARM SB, H, SH**	**ARM T, BT**	**Thumb-2 Word, B, SB, H, SH, D**	**Thumb-2 T, BT, SBT, HT, SHT**
1	offset: – 4095 to +4095	offset: –255 to +255	Not available	offset: –255 to +255 if writeback, –255 to +4095 otherwise	offset: 0 to +255, writeback not allowed
2	offset: – 4095 to +4095	offset: –255 to +255	offset: – 4095 to +4095	offset: –255 to +255	Not available
3	Full range of `{, <opsh>}`	`{, <opsh>}` not allowed	Not available	`<opsh>` restricted to `LSL #<sh>`, `<sh>` range 0 to 3	Not available
4	Full range of `{, <opsh>}`	`{, <opsh>}` not allowed	Full range of `{, <opsh>}`	Not available	Not available
5	label within +/– 4092 of current instruction	Not available	Not available	label within +/– 4092 of current instruction	Not available
6	offset: –255 to +255	-	-	offset: –1020 to +1020, must be multiple of 4.	-
7	`{, <opsh>}` not allowed	-	-	Not available	-
8	label within +/– 252 of current instruction	-	-	Not available	-
9	Rd1 even, and not r14, Rd2 == Rd1 + 1.	-	-	Rd1 != PC, Rd2 != PC	-
10	Rm1 even, and not r14, Rm2 == Rm1 + 1.	-	-	Rm1 != PC, Rm2 != PC	-

ARM and Thumb-2 Instruction Set
Quick Reference Card

Coprocessor operations	§	Assembler		Action	Notes
Data operations		`CDP{2} <copr>, <op1>, CRd, CRn, CRm{, <op2>}`		Coprocessor defined	C2
Move to ARM register from coprocessor		`MRC{2} <copr>, <op1>, Rd, CRn, CRm{, <op2>}`		Coprocessor defined	C2
Two ARM register move	5E	`MRRC <copr>, <op1>, Rd, Rn, CRm`		Coprocessor defined	
Alternative two ARM register move	6	`MRRC2 <copr>, <op1>, Rd, Rn, CRm`		Coprocessor defined	C
Move to coproc from ARM reg		`MCR{2} <copr>, <op1>, Rd, CRn, CRm{, <op2>}`		Coprocessor defined	C2
Two ARM register move	5E	`MCRR <copr>, <op1>, Rd, Rn, CRm`		Coprocessor defined	
Alternative two ARM register move	6	`MCRR2 <copr>, <op1>, Rd, Rn, CRm`		Coprocessor defined	C
Loads and stores, pre-indexed		`<op>{2} <copr>, CRd, [Rn, #+/-<offset8*4>]{!}`	op: `LDC` or `STC`. offset: multiple of 4 in range 0 to 1020.	Coprocessor defined	C2
Loads and stores, zero offset		`<op>{2} <copr>, CRd, [Rn] {, 8-bit copro. option}`	op: `LDC` or `STC`.	Coprocessor defined	C2
Loads and stores, post-indexed		`<op>{2} <copr>, CRd, [Rn], #+/-<offset8*4>`	op: `LDC` or `STC`. offset: multiple of 4 in range 0 to 1020.	Coprocessor defined	C2

Miscellaneous operations		§	Assembler	Action	Notes
Swap word			`SWP Rd, Rm, [Rn]`	temp := [Rn], [Rn] := Rm, Rd := temp.	A, D
Swap byte			`SWPB Rd, Rm, [Rn]`	temp := ZeroExtend([Rn][7:0]), [Rn][7:0] := Rm[7:0], Rd := temp	A, D
Store return state		6	`SRS{IA|IB|DA|DB} SP{!}, #<p_mode>`	[SPm] := LR, [SPm + 4] := CPSR	C, I
Return from exception		6	`RFE{IA|IB|DA|DB} Rn{!}`	PC := [Rn], CPSR := [Rn + 4]	C, I
Breakpoint		5	`BKPT <imm16>`	Prefetch abort *or* enter debug state. 16-bit bitfield encoded in instruction.	C, N
Secure Monitor Call		Z	`SMC <imm4>`	Secure Monitor Call exception. 4-bit bitfield encoded in instruction. Formerly SMI.	
Supervisor Call			`SVC <imm24>`	Supervisor Call exception. 24-bit bitfield encoded in instruction. Formerly SWI.	N
No operation		6K	`NOP`	None, might not even consume any time.	N, V
Hints	Debug Hint	7	`DBG`	Provide hint to debug and related systems.	
	Data Memory Barrier	7	`DMB`	Ensure the order of observation of memory accesses.	C
	Data Synchronization Barrier	7	`DSB`	Ensure the completion of memory accesses,	C
	Instruction Synchronization Barrier	7	`ISB`	Flush processor pipeline and branch prediction logic.	C
	Set event	6K	`SEV`	Signal event in multiprocessor system. NOP if not implemented.	N
	Wait for event	6K	`WFE`	Wait for event, IRQ, FIQ, Imprecise abort, or Debug entry request. NOP if not implemented.	N
	Wait for interrupt	6K	`WFI`	Wait for IRQ, FIQ, Imprecise abort, or Debug entry request. NOP if not implemented.	N
	Yield	6K	`YIELD`	Yield control to alternative thread. NOP if not implemented.	N

Notes			
A	Not available in Thumb state.	**P**	Rn can be the PC in Thumb state in this instruction.
B	Can be conditional in Thumb state without having to be in an IT block.	**Q**	Sets the Q flag if saturation (addition or substraction) or overflow (multiplication) occurs. Read and reset the Q flag using `MRS` and `MSR`.
C	Condition codes are not allowed in ARM state.	**R**	`<sh>` range is 1-32 in the ARM instruction.
C2	The optional 2 is available from ARMv5. It provides an alternative operation. Condition codes are not allowed for the alternative form in ARM state.	**S**	The S modifier is not available in the Thumb-2 instruction.
D	Deprecated. Use `LDREX` and `STREX` instead.	**T**	Not available in ARM state.
G	Updates the four GE flags in the CPSR based on the results of the individual operations.	**U**	Not allowed in an IT block. Condition codes not allowed in either ARM or Thumb state.
I	IA is the default, and is normally omitted.	**V**	The assembler inserts a suitable instruction if the NOP instruction is not available.
L	ARM: <imm8m>. 16-bit Thumb: multiple of 4 in range 0-1020. 32-bit Thumb: 0-4095.		
N	Some or all forms of this instruction are 16-bit (Narrow) instructions in Thumb-2 code. For details see the *Thumb 16-bit Instruction Set (UAL) Quick Reference Card.*		

ARM and Thumb-2 Instruction Set Quick Reference Card

ARM architecture versions	
n	ARM architecture version *n* and above
*n*T, *n*J	T or J variants of ARM architecture version *n* and above
5E	ARM v5E, and 6 and above
T2	All Thumb-2 versions of ARM v6 and above
6K	ARMv6K and above for ARM instructions, ARMv7 for Thumb
7MP	ARMv7 architectures that implement Multiprocessing Extensions
Z	All Security extension versions of ARMv6 and above
RM	ARMv7-R and ARMv7-M only
XS	XScale coprocessor instruction

Flexible Operand 2	
Immediate value	`#<imm8m>`
Register, optionally shifted by constant (see below)	`Rm {, <opsh>}`
Register, logical shift left by register	`Rm, LSL Rs`
Register, logical shift right by register	`Rm, LSR Rs`
Register, arithmetic shift right by register	`Rm, ASR Rs`
Register, rotate right by register	`Rm, ROR Rs`

Register, optionally shifted by constant		
(No shift)	`Rm`	Same as `Rm, LSL #0`
Logical shift left	`Rm, LSL #<shift>`	Allowed shifts 0-31
Logical shift right	`Rm, LSR #<shift>`	Allowed shifts 1-32
Arithmetic shift right	`Rm, ASR #<shift>`	Allowed shifts 1-32
Rotate right	`Rm, ROR #<shift>`	Allowed shifts 1-31
Rotate right with extend	`Rm, RRX`	

PSR fields	(use at least one suffix)	
Suffix	**Meaning**	
`c`	Control field mask byte	PSR[7:0]
`f`	Flags field mask byte	PSR[31:24]
`s`	Status field mask byte	PSR[23:16]
`x`	Extension field mask byte	PSR[15:8]

Condition Field		
Mnemonic	**Description**	**Description (VFP)**
`EQ`	Equal	Equal
`NE`	Not equal	Not equal, or unordered
`CS / HS`	Carry Set / Unsigned higher or same	Greater than or equal, or unordered
`CC / LO`	Carry Clear / Unsigned lower	Less than
`MI`	Negative	Less than
`PL`	Positive or zero	Greater than or equal, or unordered
`VS`	Overflow	Unordered (at least one NaN operand)
`VC`	No overflow	Not unordered
`HI`	Unsigned higher	Greater than, or unordered
`LS`	Unsigned lower or same	Less than or equal
`GE`	Signed greater than or equal	Greater than or equal
`LT`	Signed less than	Less than, or unordered
`GT`	Signed greater than	Greater than
`LE`	Signed less than or equal	Less than or equal, or unordered
`AL`	Always (normally omitted)	Always (normally omitted)

All ARM instructions (except those with Note C or Note U) can have any one of these condition codes after the instruction mnemonic (that is, before the first space in the instruction as shown on this card). This condition is encoded in the instruction.

All Thumb-2 instructions (except those with Note U) can have any one of these condition codes after the instruction mnemonic. This condition is encoded in a preceding IT instruction (except in the case of conditional Branch instructions). Condition codes in instructions must match those in the preceding IT instruction.

On processors without Thumb-2, the only Thumb instruction that can have a condition code is `B <label>`.

Processor Modes	
`16`	User
`17`	FIQ Fast Interrupt
`18`	IRQ Interrupt
`19`	Supervisor
`23`	Abort
`27`	Undefined
`31`	System

Prefixes for Parallel Instructions	
`S`	Signed arithmetic modulo 2^8 or 2^{16}, sets CPSR GE bits
`Q`	Signed saturating arithmetic
`SH`	Signed arithmetic, halving results
`U`	Unsigned arithmetic modulo 2^8 or 2^{16}, sets CPSR GE bits
`UQ`	Unsigned saturating arithmetic
`UH`	Unsigned arithmetic, halving results

Proprietary Notice

Document Number

ARM QRC 0001M

Change Log

Issue	Date	Change	Issue	Date	Change
A	June 1995	First Release	B	Sept 1996	Second Release
C	Nov 1998	Third Release	D	Oct 1999	Fourth Release
E	Oct 2000	Fifth Release	F	Sept 2001	Sixth Release
G	Jan 2003	Seventh Release	H	Oct 2003	Eighth Release
I	Dec 2004	Ninth Release	J	May 2005	RVCT 2.2 SP1
K	March 2006	RVCT 3.0	L	March 2007	RVCT 3.1
M	Sept 2008	RVCT 4.0			

15. Watchdog Timer (WDT)

15.1 Description

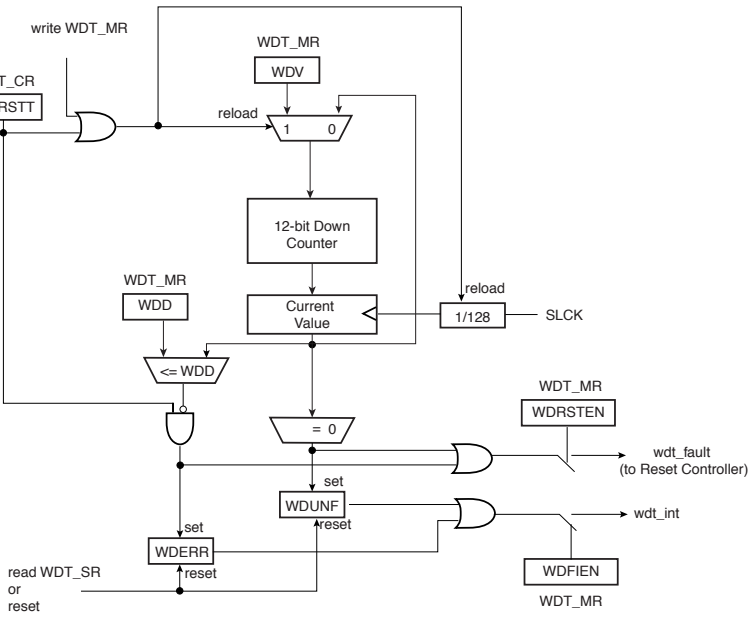
The Watchdog Timer can be used to prevent system lock-up if the software becomes trapped in a deadlock. It features a 12-bit down counter that allows a watchdog period of up to 16 seconds (slow clock at 32.768 kHz). It can generate a general reset or a processor reset only. In addition, it can be stopped while the processor is in debug mode or idle mode.

15.2 Embedded Characteristics

- 12-bit Key-protected Programmable Counter
- Provides Reset or Interrupt Signals to the System
- Counter May Be Stopped While the Processor is in Debug State or in Idle Mode
- AMBA™-compliant Interface
  - Interfaces to the ARM® Advanced Peripheral Bus

15.3 Block Diagram

Figure 15-1. Watchdog Timer Block Diagram



15.4 Functional Description

The Watchdog Timer can be used to prevent system lock-up if the software becomes trapped in a deadlock. It is supplied with VDDCORE. It restarts with initial values on processor reset.

The Watchdog is built around a 12-bit down counter, which is loaded with the value defined in the field WDV of the Mode Register (WDT_MR). The Watchdog Timer uses the Slow Clock divided by 128 to establish the maximum Watchdog period to be 16 seconds (with a typical Slow Clock of 32.768 kHz).

After a Processor Reset, the value of WDV is 0xFFF, corresponding to the maximum value of the counter with the external reset generation enabled (field WDRSTEN at 1 after a Backup Reset). This means that a default Watchdog is running at reset, i.e., at power-up. The user must either disable it (by setting the WDDIS bit in WDT_MR) if he does not expect to use it or must reprogram it to meet the maximum Watchdog period the application requires.

The Watchdog Mode Register (WDT_MR) can be written only once. Only a processor reset resets it. Writing the WDT_MR register reloads the timer with the newly programmed mode parameters.

In normal operation, the user reloads the Watchdog at regular intervals before the timer underflow occurs, by writing the Control Register (WDT_CR) with the bit WDRSTT to 1. The Watchdog counter is then immediately reloaded from WDT_MR and restarted, and the Slow Clock 128 divider is reset and restarted. The WDT_CR register is write-protected. As a result, writing WDT_CR without the correct hard-coded key has no effect. If an underflow does occur, the “wdt_fault” signal to the Reset Controller is asserted if the bit WDRSTEN is set in the Mode Register (WDT_MR). Moreover, the bit WDUNF is set in the Watchdog Status Register (WDT_SR).

To prevent a software deadlock that continuously triggers the Watchdog, the reload of the Watchdog must occur while the Watchdog counter is within a window between 0 and WDD, WDD is defined in the WatchDog Mode Register WDT_MR.

Any attempt to restart the Watchdog while the Watchdog counter is between WDV and WDD results in a Watchdog error, even if the Watchdog is disabled. The bit WDERR is updated in the WDT_SR and the “wdt_fault” signal to the Reset Controller is asserted.

Note that this feature can be disabled by programming a WDD value greater than or equal to the WDV value. In such a configuration, restarting the Watchdog Timer is permitted in the whole range [0; WDV] and does not generate an error. This is the default configuration on reset (the WDD and WDV values are equal).

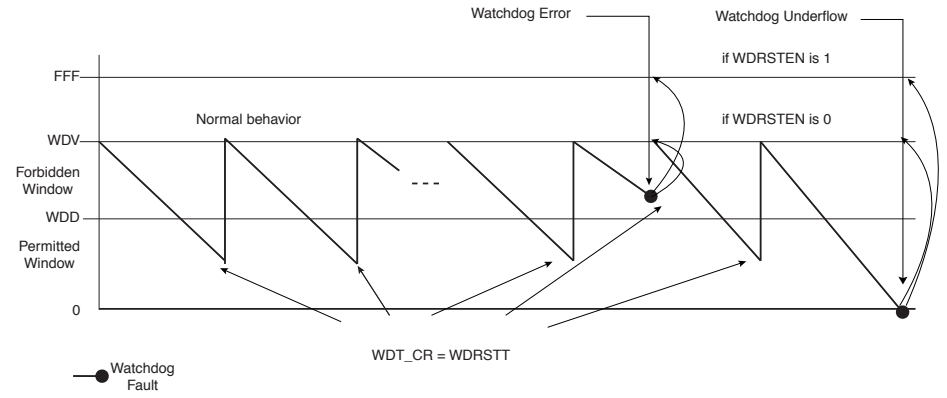
The status bits WDUNF (Watchdog Underflow) and WDERR (Watchdog Error) trigger an interrupt, provided the bit WDFIEN is set in the mode register. The signal “wdt_fault” to the reset controller causes a Watchdog reset if the WDRSTEN bit is set as already explained in the reset controller programmer datasheet. In that case, the processor and the Watchdog Timer are reset, and the WDERR and WDUNF flags are reset.

If a reset is generated or if WDT_SR is read, the status bits are reset, the interrupt is cleared, and the “wdt_fault” signal to the reset controller is deasserted.

Writing the WDT_MR reloads and restarts the down counter.

While the processor is in debug state or in idle mode, the counter may be stopped depending on the value programmed for the bits WDIDLEHLT and WDDBGHLT in the WDT_MR.

Figure 15-2. Watchdog Behavior



15.5 Watchdog Timer (WDT) User Interface

Table 15-1. Register Mapping

Offset	Register	Name	Access	Reset
0x00	Control Register	WDT_CR	Write-only	-
0x04	Mode Register	WDT_MR	Read-write Once	0x3FFF_2FFF
0x08	Status Register	WDT_SR	Read-only	0x0000_0000

15.5.1 Watchdog Timer Control Register

Name: WDT_CR

Address: 0x400E1A50

Access: Write-only

31	30	29	28	27	26	25	24
KEY							
23	22	21	20	19	18	17	16
–	–	–	–	–	–	–	–
15	14	13	12	11	10	9	8
–	–	–	–	–	–	–	–
7	6	5	4	3	2	1	0
–	–	–	–	–	–	–	WDRSTT

- **WDRSTT: Watchdog Restart**

0: No effect.

1: Restarts the Watchdog.

- **KEY: Password**

Should be written at value 0xA5. Writing any other value in this field aborts the write operation.

15.5.2 Watchdog Timer Mode Register

Name: WDT_MR

Address: 0x400E1A54

Access: Read-write Once

31	30	29	28	27	26	25	24
		WDIDLEHLT	WDDBGHLT	WDD			
23	22	21	20	19	18	17	16
WDD							
15	14	13	12	11	10	9	8
WDDIS	WDRPROC	WDRSTEN	WDFIEN	WDV			
7	6	5	4	3	2	1	0
WDV							

- **WDV: Watchdog Counter Value**

Defines the value loaded in the 12-bit Watchdog Counter.

- **WDFIEN: Watchdog Fault Interrupt Enable**

0: A Watchdog fault (underflow or error) has no effect on interrupt.

1: A Watchdog fault (underflow or error) asserts interrupt.

- **WDRSTEN: Watchdog Reset Enable**

0: A Watchdog fault (underflow or error) has no effect on the resets.

1: A Watchdog fault (underflow or error) triggers a Watchdog reset.

- **WDRPROC: Watchdog Reset Processor**

0: If WDRSTEN is 1, a Watchdog fault (underflow or error) activates all resets.

1: If WDRSTEN is 1, a Watchdog fault (underflow or error) activates the processor reset.

- **WDD: Watchdog Delta Value**

Defines the permitted range for reloading the Watchdog Timer.

If the Watchdog Timer value is less than or equal to WDD, writing WDT_CR with WDRSTT = 1 restarts the timer.

If the Watchdog Timer value is greater than WDD, writing WDT_CR with WDRSTT = 1 causes a Watchdog error.

- **WDDBGHLT: Watchdog Debug Halt**

0: The Watchdog runs when the processor is in debug state.

1: The Watchdog stops when the processor is in debug state.

- **WDIDLEHLT: Watchdog Idle Halt**

0: The Watchdog runs when the system is in idle mode.

1: The Watchdog stops when the system is in idle state.

- **WDDIS: Watchdog Disable**

0: Enables the Watchdog Timer.

1: Disables the Watchdog Timer.

15.5.3 Watchdog Timer Status Register

Name: WDT_SR

Address: 0x400E1A58

Access: Read-only

31	30	29	28	27	26	25	24
–	–	–	–	–	–	–	–
23	22	21	20	19	18	17	16
–	–	–	–	–	–	–	–
15	14	13	12	11	10	9	8
–	–	–	–	–	–	–	–
7	6	5	4	3	2	1	0
–	–	–	–	–	–	WDERR	WDUNF

- **WDUNF: Watchdog Underflow**

0: No Watchdog underflow occurred since the last read of WDT_SR.

1: At least one Watchdog underflow occurred since the last read of WDT_SR.

- **WDERR: Watchdog Error**

0: No Watchdog error occurred since the last read of WDT_SR.

1: At least one Watchdog error occurred since the last read of WDT_SR.

END OF EXAMINATION